# AtomVLA Framework

**Discrete atomic call**

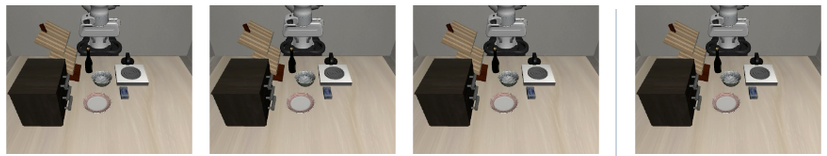
```
pick(object="bowl",
     target="above_table",
     arm="right",
     completion="held")
```

**Terminal pose**

$p^f, R^f$

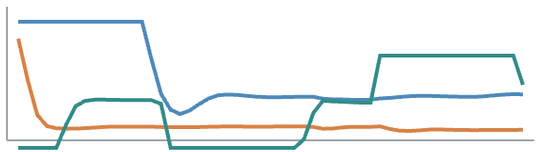
one per arm

**Primitive-specific imagined keyframes**



pregrasp | closed | lifted | move: end

**Dense action chunk**



motion + gripper commands

Q Q Q

## Program Expert

**Name + string arguments**

## Action Query

Terminal geometry

## Future Imagination

**Event keyframes replace the procedural body**

## Action Expert

**Continuous control**

Shared MoT attention [ context ; program ; Query / state ; image tokens ; action tokens ]

**VLM context $H_t$**

current RGB + instruction

**$P^{txt}$**

text CE

Q Q Q

**Learned queries**

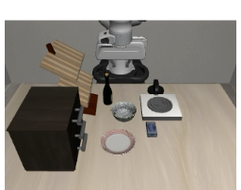
endpoint pose loss

**VAE + noisy keyframe tokens**

event roles + order + camera masks

**State prefix + noisy actions**

action flow loss



**“Put the bowl on the plate”**

Current state $s_t$

**One terminal pose**

Query predicts position and orientation

Each primitive selects its meaningful events

**Joint image / action denoising**

Action tokens read Query, state and visual tokens

Frames illustrate the representation, not model outputs